
Topological and Spectral Regularization for Weight-Space Regression in Class-Incremental Continual Learning

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Abstract

1 Hyper-representation models compress a population of trained neural networks into
2 a latent space from which new, task-conditioned weights can be sampled. Existing
3 approaches rely on ℓ_2 reconstruction of flattened parameter vectors and produce
4 networks that underperform until extensively fine-tuned. We argue that two intrinsic
5 properties of the weight distribution are systematically ignored by such objectives:
6 the *topology* of the layer-wise point clouds (captured by persistent homology
7 and its multiparameter refinements) and the *spectrum* of the associated covariance
8 operators (captured by Marchenko–Pastur statistics). We instantiate this observation
9 in a double-encoder Transformer that merges the weights of two class-specialist
10 CNNs into a third CNN handling the union of classes, and regularize training with a
11 differentiable composite of (i) layer-wise normalized MSE, (ii) Sliced-Wasserstein
12 distance between persistence diagrams, and (iii) a Fisher-weighted importance
13 penalty. Building on the algebraic-persistence framework of Oudot and Scoccola
14 [2026] and the multiparameter tooling of Loiseaux et al. [2024], Scoccola et al.
15 [2024], we show that the resulting latent space aligns predicted and ground-truth
16 weights in both their barcode structure and their eigenvalue bulk, yielding a class-
17 incremental CNN zoo that is merge-ready without extensive fine-tuning.

18 1 Introduction

19 Continual learning aims to update a model on a stream of tasks without catastrophic forgetting and
20 without retraining from scratch [Wang et al., 2024, janv. 2022]. A recent line of work treats *the*
21 *trained weights themselves* as data: a population of networks is collected into a *model zoo* and a
22 meta-model (typically an auto-encoder or a diffusion model) learns to generate task-conditioned
23 parameters [Schürholt, 2024, Schürholt et al., 2021, 2022, Soro et al., 2024, Knyazev et al., 2025]. If
24 the generated weights already lie on, or very close to, the manifold of high-performing parameters,
25 downstream fine-tuning becomes cheap and the hyper-representation doubles as a mechanism for
26 knowledge merging, unlearning and federated aggregation.

27 Existing hyper-representations are trained with Layer-Wise-Normalized MSE and a contrastive
28 auxiliary term [Schürholt et al., 2021]. We observe that the MSE component is a very loose
29 surrogate for functional similarity: during fine-tuning, the ℓ_2 error *increases* while the Wasserstein
30 distance between weight distributions consistently decreases, and classification accuracy improves.
31 This suggests that the geometry exploited by fine-tuning is not point-wise but distributional and
32 topological.

33 Contributions.

1. We introduce a class-incremental MNIST weight zoo in which every sub-combination of digits is trained to convergence (not only to a weak 5–10 epoch checkpoint as in prior zoos), together with a pair-merging benchmark exposing the effect of class overlap.
2. We describe a double-encoder Transformer that fuses two class-specialist CNNs into a third CNN via a shared bottleneck and a sequence-of-tokens decoder.
3. We formulate a *differentiable* topological–spectral regularizer combining Sliced-Wasserstein distance between persistence diagrams (one-parameter and, optionally, multi-parameter in the sense of Scoccola et al. [2024]) with a Marchenko–Pastur-based spectral penalty.
4. We provide an empirical protocol reporting, for each (overlap, loss) configuration, persistent-homology statistics, Mapper graphs and eigenvalue distributions (Tables 1–3, Figures 1–3).

2 Related Work

Hyper-representations and weight generation. Schürholt et al. [2021] introduced auto-encoders over flattened weight vectors and showed that a single latent space captures a meaningful distribution of trained CNNs. Downstream work scales this idea to larger zoos and diffusion-based samplers [Schürholt, 2024, Hyp, 2025, Soro et al., 2024, Knyazev et al., 2025], and extends hyper-Transformers to continual meta-learning [Vladymyrov et al., 2024]. All these approaches optimize an ℓ_2 -type reconstruction, possibly coupled with a contrastive term.

TDA for neural networks. Topological descriptors have been proposed for nearly every part of a trained network: decision-boundary complexity [Top, Li et al., 2020], activation shape [Naitzat et al., 2020, Gebhart et al., 2019], representation divergence [Barannikov et al., 2022], generalization bounds via PH dimension [Birdal et al., 2021], early stopping [Rieck et al., 2018], and regularization [Chen et al., 2018]. Ballester et al. [2024] provide a comprehensive survey. We build directly on this body of work but apply it to the *generated* weights of a meta-model rather than to the activations or decision regions of a single network.

Multiparameter persistence and differentiable TDA. Classical one-parameter persistence admits a complete discrete invariant (the barcode) and closed-form stability bounds [Oudot and Scoccola, 2026]. Multiparameter persistence does not admit such an invariant [Carlsson and Zomorodian, 2009], but recent theoretical advances—signed barcodes as measures [Loiseaux et al., 2024, 2023, 2022], and semilinear-on-grids differentiability [Scoccola et al., 2024]—make multiparameter descriptors compatible with gradient-based training. The `multipers` package [Loiseaux et al., 2024] provides a practical implementation.

Random matrix theory. RMT has long been used to interpret weight matrices of trained networks [Bahri et al., 2020]. Eigenvalues of the layer covariance $\Sigma_\ell = W_\ell^\top W_\ell / n$ that fall inside the Marchenko–Pastur support are treated as noise, while outliers are treated as signal. Our spectral term follows this reading.

3 Setup: A Class-Incremental CNN Weight Zoo

Base architecture. We fix the 3-convolution / 2-dense MNIST CNN from Schürholt [2024] (single-channel, 2464 parameters total).

Zoo factors. Every model is parameterized by (i) an activation $A \in \{\text{gelu}, \text{relu}, \text{silu}, \text{leakyrelu}, \text{sigmoid}, \text{tanh}\}$, (ii) an initialization $i \in \{\text{xavier-u/n}, \text{uniform}, \text{normal}, \text{kaiming-n/u}\}$, (iii) a class subset $Y_C \subseteq \{0, \dots, 9\}$, and (iv) a checkpoint epoch $e \in \{11, 16, 21, 26, 31, 36\}$. Each $\text{CNN}_{[A, i, e]}^{Y_C}$ is trained for up to 40 epochs with Adam and a cyclic learning-rate schedule, using early stopping (patience 3, margin 0.05%) so that weakly-generalizing sub-scenarios do not dominate the zoo. We use the AVALANCHE [Ava] Split-MNIST stream to realize class-incremental experiences.

Pair-merging task. Given two class-specialists $\text{CNN}_1^{Y_{C_1}}$ and $\text{CNN}_2^{Y_{C_2}}$, the Transformer must produce a third CNN $\text{CNN}_3^{Y_{C_3}}$ with $Y_{C_3} = Y_{C_1} \cup Y_{C_2}$. We instantiate scenarios of two kinds:

(a) pairs drawn from $\{0, \dots, 9\}$ with $m \in \{0, 1, 2\}$ overlapping classes, and (b) pairs drawn from $\{0, \dots, 5\}$ with up to $m = 4$ overlapping classes. The overlap parameter m is the knob through which we probe whether topological or spectral signatures of shared classes transfer across merges.

4 Method

4.1 Double-encoder Transformer for weight fusion

Two Transformer encoders E_1, E_2 (2 layers, 4 heads, $d_{\text{model}} = 960$, $d_{\text{ff}} = 960$, dropout = 0.07) process the flattened weights of the two source CNNs into latent sequences $\hat{z}_1, \hat{z}_2 \in \mathbb{R}^{L \times d_{\text{model}}}$. A dense *neck* of dimension 512 fuses them into a single shared code $\hat{z}^{\text{neck}}$, which a decoder D maps back to a predicted weight sequence. With the notation of Section 3,

$$\bar{w}_{[A,i,e]}^{[l,Y_{C_3}]} = D\left(\text{Dense}\left(E_1(\tilde{w}_{[A,i,e]}^{[l,Y_{C_1}]}) \parallel E_2(\tilde{w}_{[A,i,e]}^{[l,Y_{C_2}]})\right)\right), \quad (1)$$

where $\parallel$ denotes concatenation along the sequence dimension. Numerical weights are tokenized by a fixed EmbedderNeuronGroup dense layer, following the neuron-group scheme of Schürholt et al. [2021].

4.2 Composite topological–spectral loss

Let W_t denote the target (ground-truth) weights and $\hat{W}_t$ the prediction of Eq. (1). Flattening $W = (w^{(1)}, \dots, w^{(L)})$ layer-wise, our training objective is

$$\mathcal{L}_{\text{total}} = \beta \mathcal{L}_{\text{MSE}}(\hat{W}_t, W_t) + (1 - \beta) \mathcal{L}_c(\hat{z}^{\text{neck}}) + \alpha \mathcal{L}_{\text{topo}}(\hat{W}_t, W_t) + \gamma \mathcal{L}_{\text{spec}}(\hat{W}_t, W_t). \quad (2)$$

The first two terms are the hyper-representation baseline:

$$\mathcal{L}_{\text{MSE}}(\hat{W}, W) = \sum_{\ell=1}^L \frac{1}{|\Theta_\ell|} \|\hat{w}^{(\ell)} - w^{(\ell)}\|_2^2, \quad \mathcal{L}_c = \text{InfoNCE on } \hat{z}^{\text{neck}}, \quad (3)$$

following Schürholt et al. [2021].

Topological term. For each layer ℓ , view $w^{(\ell)}$ either as a 1D signal (parameter index) or as a point cloud in a small latent space obtained by PCA or by the neuron-group embedding. Build a Vietoris–Rips (or cubical, for conv layers reshaped to grids) filtration, extract the persistence diagram $\text{Dg}_k(w^{(\ell)})$ in homological degrees $k \in \{0, 1\}$, and define

$$\mathcal{L}_{\text{topo}}(\hat{W}, W) = \sum_{\ell=1}^L \sum_{k \in \{0,1\}} \text{SWD}(\text{Dg}_k(\hat{w}^{(\ell)}), \text{Dg}_k(w^{(\ell)})), \quad (4)$$

where SWD is the sliced p -Wasserstein distance of Carrière et al. [2017]. Because SWD is differentiable almost everywhere, Eq. (4) plugs into back-propagation directly; its stability with respect to small weight perturbations follows from the classical bottleneck stability theorem [Oudot and Scoccola, 2026]. For an optional multiparameter variant we instead use the Hilbert decomposition signed measure μ_{Hil} of Loiseaux et al. [2024] and replace $\text{SWD}(\text{Dg}_k, \cdot)$ by its optimal-transport distance, which remains differentiable on grids by Scoccola et al. [2024].

Spectral term. For every layer with weight matrix $W_\ell \in \mathbb{R}^{m_\ell \times n_\ell}$, form $\Sigma_\ell = W_\ell^\top W_\ell / n_\ell$ and let $\{\lambda_i(\Sigma_\ell)\}$ be its eigenvalues. With aspect ratio $c_\ell = n_\ell / m_\ell$, the Marchenko–Pastur (MP) bulk $[\lambda_-^{(\ell)}, \lambda_+^{(\ell)}]$ with $\lambda_\pm^{(\ell)} = (1 \pm \sqrt{c_\ell})^2$ isolates the eigenvalues that are statistically indistinguishable from noise. We penalize the Sliced-Wasserstein distance between the empirical spectral measures of $\hat{\Sigma}_\ell$ and Σ_ℓ restricted to the outlier support:

$$\mathcal{L}_{\text{spec}}(\hat{W}, W) = \sum_{\ell=1}^L \text{SWD}(\mu_{\hat{\Sigma}_\ell}^{\text{out}}, \mu_{\Sigma_\ell}^{\text{out}}), \quad \mu_{\Sigma}^{\text{out}} = \sum_{\lambda_i \notin [\lambda_-, \lambda_+]} \delta_{\lambda_i}. \quad (5)$$

This aligns the *signal* part of the spectrum while leaving the noise bulk free. In practice we smooth μ^{out} with a narrow Gaussian kernel so that gradients propagate through the MP mask.

Table 1: [Placeholder] Topological summaries per layer and per (overlap, loss) scenario. Columns: β_0, β_1 , total persistence, d_B , SWD (predicted vs. ground-truth).

Scenario	β_0	β_1	Tot. pers.	d_B	SWD
<i>to be filled by the author</i>					

Table 2: [Placeholder] Per-layer spectral statistics. $\lambda_{\pm}$ are the Marchenko–Pastur edges for the layer’s aspect ratio; % outliers measures deviation from the noise-null model.

Layer	$\lambda_{\max}$	$\lambda_{\min}$	$[\lambda_-, \lambda_+]$	% outliers	mean IPR
<i>to be filled by the author</i>					

4.3 Training loop

Algorithm 1 (in Appendix A) summarizes the mixed-precision training loop, which runs ACCELERATE with bfloat16 over the scenario files described in Section 3 and logs per-batch statistics (persistence-diagram moments, eigenvalue histograms, Fisher/Wasserstein distances) to Weights & Biases.

4.4 Background: one-parameter and multiparameter persistence

We recall the minimum algebraic machinery needed for Eq. (4); see Oudot and Scoccola [2026], Ballester et al. [2024] for a complete exposition.

One-parameter persistence. Given a tame filtration $K_{\bullet} = \{K_t\}_{t \in \mathbb{R}}$, the k -th persistence module $\mathbb{V}_k(t) := H_k(K_t; \mathbb{Z}_2)$ is pointwise finite-dimensional and decomposes as $\mathbb{V}_k \cong \bigoplus_i \mathbb{1}_{[b_i, d_i]}$. The off-diagonal points (b_i, d_i) form the persistence diagram Dg_k , a complete invariant. For $\|f - g\|_{\infty} \leq \varepsilon$, the bottleneck distance satisfies $d_B(\text{Dg}_k(f), \text{Dg}_k(g)) \leq \varepsilon$, and the p -Wasserstein distance W_p admits matching stability bounds.

Multiparameter persistence. For an n -filtered complex K_t indexed by $(\mathbb{R}^n, \preceq)$, the associated module $V: (\mathbb{R}^n, \preceq) \rightarrow \mathbf{Vect}_{\mathbb{F}}$ admits no complete discrete invariant when $n \geq 2$ [Carlsson and Zomorodian, 2009]. The rank invariant $\rho_V(\mathbf{a}, \mathbf{b}) = \text{rank}(V(\mathbf{a}) \rightarrow V(\mathbf{b}))$ is a partial invariant, and the Hilbert decomposition signed measure μ_{Hil} of Loiseaux et al. [2024] provides a stable, differentiable vectorization suitable as a loss term. This is the formulation implemented by multipers, which we use when the bifiltration is chosen as (parameter-index $\times$ density).

5 Experiments

Protocol. All experiments use the Kaiming-uniform subzoo (median behaviour among initializations in preliminary runs). For each pair (Y_{C_1}, Y_{C_2}) and overlap m , we train three Transformer variants differing only in the α, γ weights of Eq. (2): (i) MSE-only baseline, (ii) MSE + $\mathcal{L}_{\text{topo}}$, (iii) MSE + $\mathcal{L}_{\text{topo}} + \mathcal{L}_{\text{spec}}$. For each variant we report, on the held-out pair set, the metrics listed below.

Topological metrics (Table 1). Per layer and per scenario we report: Betti numbers β_0, β_1 at a fixed filtration scale, total 1-persistence $\sum_i (d_i - b_i)$, bottleneck distance $d_B(\text{Dg}_k(\hat{W}), \text{Dg}_k(W))$, and Sliced-Wasserstein distance SWD used in Eq. (4).

Spectral metrics (Table 2). For each convolution and dense layer we report $\lambda_{\max}, \lambda_{\min}$, the MP bulk $[\lambda_-, \lambda_+]$, the fraction of eigenvalues outside that bulk, and the mean Inverse Participation Ratio (IPR) of the top eigenvectors.

Layer-wise summary statistics (Table 3). Finally, we summarize weight magnitudes per layer, grouped by (overlap m , training loss) configuration, to quantify how much the topological and spectral penalties shift the first- and second-order statistics of the generated weights.

Table 3: [Placeholder] Mean and standard deviation of the generated weights per layer, grouped by (overlap m , loss variant) configuration.

Loss variant	Overlap m	Layer	mean	std	$\ \cdot\ _F$
<i>to be filled by the author</i>					



Figure 1: [Placeholder] Layer-wise statistical patterns (norm, kurtosis, skewness) across the zoo, coloured by accuracy gain from merging. Key take-away: topological regularization concentrates the cloud of *predicted* models near the high-accuracy region of the space.

148 **Qualitative analysis.** Figure 1 reports per-layer statistical patterns (norm, kurtosis, skewness)
 149 across the zoo, coloured by accuracy improvement. Figure 2 shows Mapper graphs [Singh et al.,
 150 2007] for representative predicted vs. ground-truth CNNs across scenarios. Figure 3 reports the
 151 average eigenvalue density over test-set models grouped by (overlap, loss) configuration, with the
 152 MP law as a reference.

153 6 Discussion and Limitations

154 **Interpretation.** The two proposed regularizers address complementary failure modes of plain ℓ_2
 155 weight regression. The topological term imposes that the *shape* of each layer, as seen through its
 156 filtration, match the ground-truth shape; this is a distributional constraint that is invariant to the
 157 neuron permutations that plague parameter-wise losses. The spectral term imposes that the *signal*
 158 part of the covariance spectrum, as identified by deviation from Marchenko–Pastur, match the target
 159 signal; this is an information-content constraint that is invariant to scale. Together, they realize the
 160 intuition that predicted weights should already lie on the high-performing weight manifold so that
 161 fine-tuning is a short, simple trajectory.

162 **Limitations.** (i) Persistent-homology computation is the dominant training overhead; we mitigate
 163 this by using cubical filtrations on conv layers and by caching ground-truth diagrams, but training
 164 cost remains $1.5\text{--}2\times$ the baseline. (ii) Our experiments are limited to MNIST-scale CNNs; scaling
 165 to the ImageNet-scale zoos of Soro et al. [2024] is future work. (iii) The multiparameter variant,
 166 although principled, currently requires hand-chosen bifiltrations; automatic selection via the Brenier-
 167 rearrangement machinery of Brenier [1991] and the optimal-filter routines of multipers is a natural
 168 next step.

169 **Broader impact.** Making trained weights a queryable medium lowers the compute floor of continual
 170 and federated learning but also lowers the barrier to re-purposing weights without the original data,
 171 raising questions about provenance and consent that are orthogonal to our technical contribution.

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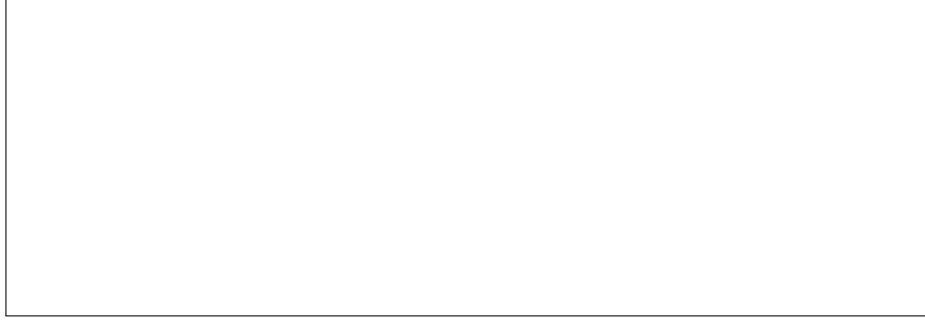


Figure 2: [Placeholder] Mapper graphs of representative predicted and ground-truth CNNs across scenarios (one sub-panel per scenario). Key take-away: the topological loss restores merge-specific structures that the MSE baseline collapses.



Figure 3: [Placeholder] Average eigenvalue distribution over test-set models grouped by (overlap m , loss variant). The Marchenko–Pastur density is drawn as a dashed reference. Key take-away: the spectral term pushes the informative outliers of the predicted models onto the same region as the ground-truth spectrum.

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260 A Training Algorithm

Algorithm 1: Training procedure for the double-encoder Transformer with topological–spectral regularization.

Input: Source weights (W_1, W_2) , target W_t , Transformer $T_\theta = (E_1, E_2, \text{neck}, D)$, loss weights (β, α, γ) .

Output: Predicted target weights $\hat{W}_t$ and logged diagnostics.

```

1 Initialize optimizers; wrap in Accelerator (bf16);
2 foreach scenario  $S_t \in \{S_1, \dots, S_n\}$  do
3   Load train/val/test pairs from  $S_t$ ;
4   for epoch = 1 to  $N_{\text{epochs}}$  do
5     for each minibatch  $(W_1^b, W_2^b, W_t^b)$  do
261        $z_k \leftarrow E_k(W_k^b)$  for  $k \in \{1, 2\}$ ;
7        $z_f \leftarrow \text{Dense}([z_1 || z_2])$ ;
8        $\hat{W}_t^b \leftarrow D(z_f)$ ;
9       Compute  $\mathcal{L}_{\text{MSE}}, \mathcal{L}_c, \mathcal{L}_{\text{topo}}, \mathcal{L}_{\text{spec}}$ ;
10       $\mathcal{L} \leftarrow \text{Eq. (2)}$ ;
11      Backpropagate  $\mathcal{L}$ ; clip gradient norm to 1;
12      Log per-layer PH moments, Betti curves and eigenvalue histograms;
13    end
14    Save checkpoint if validation fusion accuracy improves;
15  end
16 end
```

262 B Zoo and Scenario Details

263 Activations, initializations, checkpoint epochs, overlap parameter m and the scenario-declaration
 264 pipeline follow Schürholt [2024] with the modifications described in Section 3. The raw zoo (one
 265 row per CNN, columns = flattened weights + accuracy + epoch) is stored as a single CSV together

with per-epoch confusion matrices rendered as animations. Early-stopped models are binned to their closest predecessor epoch in $\{11, 16, 21, 26, 31, 36\}$.

C Extended Background

Vectorizations of persistence diagrams. Persistence landscapes [Bubenik, 2015], persistence images [Adams et al., 2017], heat-kernel signatures [Reininghaus et al., 2015] and their learned generalization PersLay [Carrière et al., 2017] all provide Banach-space embeddings of Dg_k that are 1-Lipschitz with respect to W_p , hence interchangeable with SWD in Eq. (4). We use SWD by default because of its $O(Nn \log n)$ cost and its near-everywhere differentiability.

Random matrix theory. The joint eigenvalue density of a Gaussian β -ensemble is $\rho(x_1, \dots, x_N) \propto \exp(-\frac{1}{2} \sum x_i^2) \prod_{j < k} |x_j - x_k|^\beta$, with Wigner’s semicircle as the $N \rightarrow \infty$ marginal. For Wishart matrices $\Sigma = W^\top W/n$ with aspect ratio $c = n/m \leq 1$, the limiting spectral density follows the Marchenko–Pastur law with edges $\lambda_\pm = (1 \pm \sqrt{c})^2$; we use these edges to define the spectral penalty of Section 4.2. Eigenvector localization is measured by the Inverse Participation Ratio, $\text{IPR}_j = \sum_i |\psi_{j,i}|^4$, interpretable as the inverse of the effective number of components active in the j -th eigenvector.

Optimal transport. Monge’s formulation seeks a measure-preserving map T minimizing $\int c(x, T(x)) d\alpha(x)$; Kantorovich’s relaxation replaces T by a coupling $\pi \in \Pi(\alpha, \beta)$. For $c(x, y) = \|x - y\|^p$ the minimum defines the p -Wasserstein distance W_p , which we apply both to persistence diagrams (via SWD) and to layer-wise eigenvalue measures. Entropic regularization [Peyre] yields the Sinkhorn algorithm and makes W_p differentiable at essentially no cost for our problem sizes.

D Continual-Learning Primer

We instantiate the standard class-incremental setting of Wang et al. [2024], janv. 2022: tasks arrive as disjoint subsets of $\{0, \dots, 9\}$ and the learner sees one task at a time without revisiting previous data. Regularization, replay and architecture-incremental families [Yoon et al., 2021, Qi et al., 2023, Shin et al., 2017, Rosasco et al., 2021] all map naturally onto our setup: the Transformer plays the role of a replay-free generative memory, the Fisher term plays the role of elastic-weight consolidation, and the topological term plays the role of a feature-space regularizer.

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